

Université de Bourgogne Franche-Comté
Master Behavioural Ecology and Wildlife Management
Bibliographic report, 2nd year Master program BEWM 2019-2020

Social factors and patterns of senescence in mammals

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within the Myanmar Timber Elephant Project

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1. INTRODUCTION

1.1. CONTEXT AND DEFINITIONS

Senescence and ageing are processes experienced by virtually all living organisms (Nussey et al., 2013). However, their definitions are not as straightforward as it appears and numerous papers use senescence and ageing as synonymous for the sake of simplicity (*e.g.* Fabian and Flatt, 2011; Flatt and Partridge, 2018; Lemaître et al., 2015; Rose, 1991). Thus, an evolutionary definition of ageing and senescence would be presented as an inevitable and irreversible age-dependent decrease in reproductive success and increase mortality at the demographic level, resulting from a deterioration of the physiological function with advanced-age. Withal, in the present review we will mark a clear distinction between ageing and senescence in the way Monaghan et al. (2008) did. Senescence will here be defined as an age-related change in a trait value associated with detrimental changes in reproductive fitness and/or survival (loss of function or impaired performance) and ageing is defined as an age-related change in trait-value with no consequences on survival and/or reproduction abilities (Monaghan et al., 2008), although defining a trait as ageing or senescing might be context-dependent. Now that the distinction between senescence and ageing has been pointed out, note that we will use the term ‘senescence’ whenever we talk about theory but we might use ‘ageing’ when dealing with examples. Indeed, consequences of age-related changes in a trait value on individual fitness is rarely known.

Until the 1950’s, the paradigm was that senescence was a feature of humans and captive, domestic and laboratory animals. Indeed, the early idea was that senescence could not occur in the wild because of high extrinsic mortality (Nussey et al., 2013). As an example, Lack (1943) noted that the mean age at death in wild blackbirds (*Turdus merula*) was 2.5 times shorter than what previously found in captive blackbirds, and “that hardly any wild Blackbirds ever attain[ed] to ‘old age’”. Then, thanks to the works of Haldane in 1941, Medawar in 1952 and Williams in 1957 – based on the idea that the strength of selection weakens with age – and Hamilton in 1966 predicting that senescence should start at sexual maturity, there was no reason for wild organisms not to senesce, even without showing infirmity or frailty. While the prediction that senescence should start at sexual maturity has been strongly questioned (see Gaillard and Lemaître, 2017; Péron et al., 2010; Tidière et al., 2015), this idea has participated in disentangling the concepts of senility and senescence (Nussey et al., 2013). Indeed, comparative studies between captive and wild populations of a same species have shown that both senesced, although in different ways (Hämäläinen et al., 2014; Tidière et al., 2016). Zoos captive mammals lived longer than their free-ranging conspecifics – particularly for short-lived

species – with wild individuals tending to experience earlier onset of actuarial senescence (Tidière et al., 2016). Additionally, this study showed that the onset of senescence for both wild and captive female lions (*Panthera leo*) was delayed relatively to age at sexual maturity.

Senescence has now been studied in virtually the entirety of the tree of life, from mammals (Promislow, 1991), birds (Travin & Feniouk, 2016), fish (Reznick et al., 2002), reptiles (Massot et al., 2011), insects (Keller & Genoud, 1997) or plants (Chmilar & Laird, 2019) to unicellular species in certain conditions (Florea, 2017). Jones et al., (2014) demonstrated that senescence was almost universal, but was also highly diverse and that senescence patterns had to be placed on a continuum ranging from strong age-dependent deterioration up to age-dependent improvement in some rarer cases. The author observed some species with postreproductive lifespans as in humans (*Homo sapiens*) or killer whales (*Orcinus orca*), some species with decreasing mortality with age as the desert tortoise (*Gopherus agassizii*), or others with constant rates of mortality and fecundity throughout their lives and extremely long longevity, like *Hydra magnipapillata*, living up to 1,400 years old.

So, in the last decades, senescence raised a strong interest within scientific research. Indeed, the question of senescence and ageing in humans and whether we have to try and fight it led us to study (bio)gerontology at first in the laboratory rather than in the wild, in order to elucidate the mechanisms of getting old (Monaghan et al., 2008) – and because wild animals were thought not to senesce because of high extrinsic mortality (Lemaître et al., 2015). In that respect, papers dealing with senescence and ageing in humans drastically increased during the last three decades (e.g. Hayflick, 2000; Hill and Kaplan, 1999; Viña et al., 2018). But if the study of senescence and ageing presents such a vivid interest, it is also because of the paradox these processes imply. How could natural selection, designing organisms in a way to maximise the survival and/or reproductive success (i.e. Darwinian fitness) of an individual allow the evolution of processes diminishing performances and contribution to the next generation, instead of favouring an immortal life (Fabian & Flatt, 2011; Partridge & Barton, 1993)?

1.2. THE CLASSICAL EVOLUTIONARY THEORY OF AGEING

The first attempts to solve this evolutionary paradox can be traced back to Haldane's and Medawar's idea, in the 1940-50's, that the strength of natural selection weakens with age according to the Fisher's reproductive value (i.e. the contribution of an individual to the next generation). The use of the fisherian reproductive value to predict the strength of selection would then be refined by Hamilton (1966). The latter focused more on the contribution of the whole gene pool of a population to the next generations, rather than on an individual's

contribution to it (Charlesworth, 2000; Monaghan et al., 2008; Partridge & Barton, 1993). From this idea that the strength of the selection decrease with old ages, Medawar developed the mutation accumulation theory of ageing in his 1952's book, *An Unsolved Problem in Biology*. This theory posits that at the population level, very few individuals would survive to old age because of extrinsic mortality. Therefore, mutations with deleterious effects at old age could not be sorted out from the gene pool of the population and should subsequently accumulate. In that respect, senescence could be considered as a by-product of the natural selection (Nussey et al., 2013).

In 1957, George C. Williams developed an idea mentioned by Medawar to formulate his antagonistic pleiotropy theory of ageing. This idea was that natural selection should favour the selection of beneficial early-acting genes, rather than the sorting of detrimental late-acting genes, because the magnitude of the effect of an allele on an individual's fitness is compensated by the remaining probability of reproduction, decreasing with age. It results that a small advantage of an allele at young ages should outweigh a strong disadvantage of this allele at old ages. The antagonistic pleiotropy theory does not consider senescence as a by-product of natural selection anymore as senescence is actually actively favoured by selection (Williams, 1957). This theory implies that senescence is shaped by a trade-off between early benefits and late costs.

A third theory, the disposable soma theory (Kirkwood, 1977), also based its statement on a trade-off: the trade-off involved in the resource allocation towards reproduction, growth and somatic maintenance. The disposable soma theory posits that organisms accumulate resources from their environment and use these resources to different metabolic tasks. The problem is that an allocated amount of resource cannot be used again for other purposes. It results from this that different strategies in how to allocate resources can be adopted, and the question is therefore to determine the optimal strategy maximising fitness. It appears that, given the existence of extrinsic mortality over lifetime, allocation towards somatic maintenance is optimal in fitness output in proportion incompatible with a non-senescing life (Kirkwood & Rose, 1991).

In the years following Williams' antagonistic theory, the focus on the antagonistic pleiotropy between early and late effects has shifted towards the antagonistic pleiotropy between survival and reproduction. Antagonistic pleiotropy and disposable soma theories are therefore quite similar theories, capturing the same concept at different physiological levels and using different approaches to answer the same question. The former adopts a population genetics approach, the latter an optimisation theory approach focused on the physiology of senescence and the random

molecular damages associated (Kirkwood, 1977; Kirkwood & Rose, 1991). While the disposable soma theory can easily be demonstrated as long as trade-offs between say growth early in life and a fitness-related traits later in life are evidenced, the antagonistic pleiotropy requires harder work and quantitative genetics tools to assess whether an allele provides early-life benefits and late-life costs and few work did so (Nussey et al., 2008a).

2. TOWARDS A SOCIAL THEORY OF AGEING

Although this classical evolutionary theory of ageing seemed to have solved the Darwinian paradox of senescence – the Medawar’s unsolved problem in biology (Holliday, 2006) – and has allowed us to observe senescence in wild organisms, there is a huge diversity in patterns of senescence (Jones et al., 2014). Among this diversity of senescence patterns, some species show long postreproductive lifespans as in humans where women (Canadian and Finnish born between the eighteenth and nineteenth century) experience menopause around 49 years old, while their life expectancy is around 70 years old (Lahdenperä et al., 2004) or resident killer whales in which females cease reproduction around 40 years old but can live up to 90 years old, being non-reproductive for more than half their lifespan (Croft et al., 2015). We could also think about eusocial insects or mammals including non-reproductive casts like the naked mole-rats (*Heterocephalus glaber*) in which both reproductive and non-reproductive individuals could frequently live more than 20 years in captivity (Sherman & Jarvis, 2002), or even cooperative breeding species of birds, fish and mammals, in which subordinates experience long non-reproductive lifespans, providing help to breeding individuals in rearing young before to reproduce themselves (Koenig et al., 1992). All of these species with seemingly counterintuitive senescence patterns are social species. This discrepancy between what theories predict and what is observed cannot be explained solely by the classical theories on ageing. Therefore, a growing body of research is trying to complete and refine these theories by adding the concept of sociality into them (e.g. Bourke, 2007; Lee, 2003; Lucas & Keller, 2020).

2.1. SOCIALITY

Sociality is defined by the tendency of conspecific individuals to live in groups (Alexander, 1974; Clutton-Brock, 2016). A group is characterized by a social system defined by three components: the social organisation, the social structure and the mating system (Kappeler & van Schaik, 2002). In other words, a social system is characterized by the composition of a group (group size, sexual composition) and how members of the group interact together, considering sexual interactions as a subset of social interactions. Based on the observation that

female reproductive fitness was decreased with increasing group size when only maternal cares were provided to offspring, but that their reproductive fitness was positively correlated with group size when allomaternal care (*i.e.* care to offspring by other individuals than the mother, including paternal care) were provided, Silk (2007) highlighted the relevance of placing social systems upon a continuum of care to offspring, from the absence of parental care to singular breeding (*i.e.* a unique dominant female monopolises the reproduction, made possible by the presence of non-breeding helpers). Furthermore, living in group provides individuals with gains of fitness through protection against predation, foraging efficiency or mating opportunities, but is also costly in terms of competition to access to different resources or diseases transmission (Alexander, 1974; Clutton-Brock, 2016; Silk, 2007). However, all groups of individuals are not social groups, for example when individuals gather because of environmental constraints, with low interactions between them, *i.e.* gregarism.

Sociality seems to be able to provide insights to explain the diversity of patterns of senescence that we can observe both across and within species. Bourke (2007) reviewed studies comparing between- or within-species differences in senescence. For example, female odontocete whales show long postreproductive lives while female mysticete whales do not. It appears that the former also exhibit life in kin groups and maternal care, while the latter do not (Bourke, 2007). When comparing species of anthropoid primates, we observe that as the ratio of female/male survival decreases among species, paternal investment in care to offspring increases (Allman et al., 1998). Therefore, we could observe an association between the diversity of patterns of senescence and the diversity of social systems. Based on these observations, the classical theories of ageing would need indeed to be complemented and extended with the inclusion of social factors, providing a broader social theory of ageing, necessary to better account for patterns of senescence in humans (Hill & Kaplan, 1999).

2.2. KIN SELECTION

2.2.1. *The inclusion of kin selection in the classical theories of ageing*

One of the nine predictions of Williams (1957) states that “there should be little or no postreproductive period in the normal life-cycle of any species [...] as sterility is the selective equivalent of death”. While classical theories predict that longer lifespans should not be selected if not accompanied with reproduction (Hamilton, 1966; Lahdenperä et al., 2004; Williams, 1957), we do observe species in which individuals cease reproduction in favour of the reproduction of other members of the family group. One common features of these species

with seemingly counterintuitive life patterns is that all of them are social species. This suggests that sociality, influencing longevity, can influence senescence. How does sociality promote the evolution of long postreproductive lifespans or the existence of subordinate individuals sacrificing their own reproduction for the reproduction of dominant individuals?

Since Hamilton (1963) proposed a solution to solve the problem of the evolution of altruistic behaviours in animals and that the level of selection was deeply anchored within the scientific community as being gene centered by Dawkins (1976), the concept of inclusive fitness (*i.e.* direct fitness and indirect fitness through kin) had become a paradigm in evolutionary and behavioural ecology. This kin selection theory posits that individuals may present altruistic behaviours if the costs c of performing the act for the donor is outweighed by the benefits b the act implies for the recipient times the relatedness r between the two individuals, *i.e.* $r \cdot b - c > 0$ (Hamilton, 1963). In other words, altruistic behaviours may emerge if helping an individual sharing part of one's own genes allows him to pass on his genes. As patterns of senescence are to be linked with individual fitness, they are to be linked with inclusive fitness, resulting from cooperation with related individuals¹ (Bourke, 2007; Lucas & Keller, 2020, see also Box 1).

2.2.2. *Kin selection promotes survival of juveniles*

Killer whales experience menopause and long postreproductive life, contrary to what is predicted by the classical theory of ageing. This postreproductive life allows grandmother killer whales to provide care to grandcalves, enhancing their survival and thus providing these grandmother with gains in inclusive fitness (Nattrass et al., 2019), the so-called 'grandmother effect' (Hawkes et al., 1998). Similarly, human females with more grandchildren had longer postreproductive lifespans than females with fewer grandchildren, suggesting that grandmother effects promote longer lifespans to allow grandmother to provide care to grandchildren and decrease investment of daughter in raising children (Lahdenperä et al., 2004). These postreproductive lifespans can be more generally explained by intergenerational transfers. These intergenerational transfers include both parental care and grandparental care. The idea is that it exists trade-offs between fertility (*i.e.* the number of offspring) and investment per offspring (intergenerational transfers). In the case where no investment towards offspring is made after birth, patterns of senescence are similar to what the classical theory predicts. On the

¹ The classical evolutionary theories of ageing we described above are based upon the idea that the strength of the natural selection weakens with increasing age. In that respect, theoreticians tried to explain the evolution of senescence by justifying senescence as a way of maximising the individual fitness. As we might expect natural selection and kin selection to act on patterns of senescence, it is likely that sexual selection and mating systems, which are directly linked with social systems, also participate in shaping senescence patterns (Box 1).

other hand, when intergenerational transfers occur, juveniles mortality is predicted to decrease with age, due to parental care for example, and postreproductive lifespans to increase in the caregiver because of investments to others (Lee, 2003).

2.2.3. Kin selection and reproductive costs

In cooperatively breeding societies, dominant breeding individuals are helped by non-breeding (or irregularly breeding) subordinates called helpers, generally kin, – but not always – to rear young (Clutton-Brock, 2016; Koenig et al., 1992). In such social level, reproductive individuals helped by non-reproductive ones gain benefits in survival, reproductive lifespan and ultimately in lifetime reproductive success because of a decrease in reproductive costs – *i.e.* the load-lightening hypothesis (Crick, 1992). Breeding female meerkats (*Suricata suricatta*) exhibited lower daily energetic expenditures with increasing number of helpers (Scantlebury et al., 2002),

Box 1. Theory of ageing and sexual selection

In mammals, males tend to experience shorter lifespans and higher rates of senescence than females, for example due to the investment in secondary sexual costly traits² or intrasexual competition (Bonduriansky et al., 2008; Maklakov & Lummaa, 2013; Tidière et al., 2015), as it is the case in European badgers (*Meles meles*) in which males experience stronger body mass ageing than females due to stronger intrasexual competition (Beirne et al., 2015). Also, the existence of sexual conflicts between males and females (*i.e.* genetic co-evolution of males and females with contradictory interests) shapes life histories of the other sex (Bonduriansky et al., 2008). Whenever both sexes are observed, sex-differences arise in their patterns of senescence (Berger et al., 2018; Briga et al., 2019; Carranza & Pérez-Barbería, 2007; Hämäläinen et al., 2014; Maklakov & Lummaa, 2013). Indeed, considering that males and females should differ in their life reproductive strategies, we might expect sex-differences in optimal life-history strategies. As reproductive strategies are strongly related to the pace of life of individuals, sexual selection, should shape senescence patterns through the trade-off in allocation towards reproduction, growth and somatic maintenance. Surprisingly, sexual selection has not been extensively studied within the framework of senescence. Lemaître et al. (2015) highlighted in their review that the great majority of papers studying reproductive senescence focused on females, probably because quantifying reproductive effort is easier than in males. Early classical theories of ageing needs to also include the potential link between sexual selection and senescence to fully apprehend it (Lemaître et al., 2015).

² Let us think about the Handicap principle (Zahavi, 1975) to emphasise potential similarities between sexual selection and senescence. Expressing a costly trait having beneficial effects at the reproductive period of life can be positively selected even if the trait is actually more costly than advantageous at older ages, reminding us of the antagonistic pleiotropy theory of ageing.

and the number of helpers during the previous breeding event was positively correlated with the mother's weight at the following breeding (Russell et al., 2003). *Allocare* (*i.e.* care to offspring other than by the female) can also be provided by males. Comparing around 500 species of mammals, West and Capellini (2016) found that females of species with male care had shorter lactation periods and more frequent and larger litters, evidencing a decrease in reproductive costs for the female through male care.

2.2.4. *When kin selection generates diversity in patterns of senescence*

Some spatially structured models found that dispersal patterns of a population could explain the evolution of longer lifespans through kin interactions (Ronce & Promislow, 2010; Travis, 2004). Indeed, in a population with low dispersal, the neighbours of an individual are likely to be relatives. As the strength of selection decreases with age, hence individual fitness too, it would then be more beneficial for the focal individual to die earlier, as the patch it occupied would be free for one of its relative, providing it with inclusive fitness gains. However, with more global dispersal patterns, the likeliness for neighbours to be relatives is decreased, so the gain in inclusive fitness when dying and freeing a patch is low selecting for longer lifespans (Ronce & Promislow, 2010; Travis, 2004). Ronce and Promislow (2010) found the same results on shorter lifespan under limited dispersal but focused their model on competition between kin (parent-offspring and between siblings). This highlights the costs of sociality through competition that also exists even between relatives (West et al., 2002) and this competition between group members can also affect patterns of senescence. Female meerkats competing for dominant position, usually between relatives, showed higher rates of ageing and lower reproductive success (Sharp & Clutton-Brock, 2011).

2.3. GROUP-LIVING AND EXTRINSIC MORTALITY

One other important aspect of sociality is the impact that group-living might have on extrinsic mortality (Lucas & Keller, 2020). A population experiencing high rates of extrinsic mortality will consequently experience high decrease in strength of selection with age, as few individuals are able to reach old ages. Thus, there is lower selective pressure to sort out deleterious mutation, or to select against alleles with detrimental effects with increased age. According to the classical theories of ageing, high extrinsic mortality should result in shorter lifespans and higher rates of senescence (Gaillard & Lemaître, 2017; Ricklefs, 2010; Williams, 1957 but see Maklakov, 2013 for a discussion about positive, rather than antagonistic, pleiotropy). It has been shown

that species with greater abilities to evade predation risk, natural hazards or disease transmission presented greater longevity and delayed senescence. For example, volant species show delayed senescence and increased longevity compared to non-volant species of a same body size. Bats, the only volant mammals, experience lower extrinsic mortality and longer lifespans than their terrestrial relatives (Austad & Fischer, 1991; Shattuck & Williams, 2010). In the same idea, arboreal mammals have longer longevity than terrestrial mammals for similar body sizes (Shattuck & Williams, 2010; Williams & Shattuck, 2015), as well as for fossorial species (Healy et al., 2014). However, a comparison of fossorial versus terrestrial mammals, whether they were eusocial or not, evidenced that fossoriality had not the primary role in explaining longevity, while body mass had, followed by eusociality (Williams & Shattuck, 2015). Indeed, among the costs and benefits of social life, group living is generally thought to provide individual protection against predation (Hamilton, 1971). For example, the white-nosed coatis (*Nasua narica*) experienced lower predation rates when living in groups than solitary individuals (Hass & Valenzuela, 2002). The distance from cover at which the Alaskan moose (*Alces alces gigas*) foraged was positively correlated to group size, suggesting that group-living provides protection against predation (Molvar & Bowyer, 1994). Group-living also presents costs to individuals by enhancing within-group competition to access resources or by increasing infections and diseases transmission. The foraging efficiency of the Alaskan moose was for example decreased with group size in this species (Molvar & Bowyer, 1994). It might therefore be difficult to observe the direct effects of group-living on extrinsic mortality, as benefits and costs may compensate each other.

2.4. COEVOLUTION BETWEEN SOCIALITY AND SENESCENCE

We have seen that sociality may promote greater longevity through kin selection processes and benefits of group living on extrinsic mortality, but it is important to note that the contrary may also apply. Ross et al. (2015) formulated a model of a spatially structured population of individuals interacting according to the classical prisoner's dilemma rules. By modifying the duration of different life history stages (pre- and post-reproduction and reproduction), results suggested that longer postreproductive periods favoured cooperation, while longer pre-reproductive periods the reverse. In long-lived species, with slow biological times (*i.e.* slow pace of life), in which long periods of parental care are necessary, we might expect selection towards longer lifespans. A complex relationship of causalities exists between longevity affected by sociality and affecting sociality through proximity with relatives, sociality affecting pace of life and extrinsic mortality both acting on longevity (Figure 1 in Lucas & Keller, 2020).

3. THE LINK BETWEEN SOCIALITY AND SENESCENCE

3.1. AT THE INTERSPECIFIC LEVEL

The larger a species, the slower its life history, as large-bodied species require time and energy for growth early in life, thus delaying the age of maturity, of reproduction, number of offspring per reproduction, *etc.* (Armitage, 1981; Oli, 2004). Jones et al. (2008) evidenced that species with slower life histories, *i.e.* delayed age of primiparity, longer generation time, fewer offspring per reproduction, were the species with lower rates of senescence. They evidenced that the species pace of life was a major predictor of the magnitude of senescence. Additionally, there is evidence for an interspecific association between sociality and brain size, corrected for body size (*i.e.* relative brain size) and that larger relative brain size was associated with slower rates of senescence (Promislow, 1991). It is thus, logical to observe strong social behaviours within large species, especially in mammals³. At least, we observe that the biggest species tend to exhibit more complex social behaviours as in humans or cetaceans, both even described as eusocial species because of the existence of menopause, resulting in a within-individual division of labour and because of grandmother effects (Foster & Ratnieks, 2005; McAuliffe & Whitehead, 2005; Natrass et al., 2019). All of that suggest that senescence and sociality covary in a pace of life syndrome supporting the idea of coevolution between these two traits.

However, evidence for an evolutionary link between sociality and longevity when studying interspecific comparisons are scarce. In their review, Lucas and Keller (2020) found no correlation between longevity and group size in mammals, except in a study on a hundred species of bovids in which group size and longevity were positively correlated (Bro-Jørgensen, 2012). Williams and Shattuck (2015) also found a positive correlation between longevity and eusociality and cooperative breeding in mammals, but it is likely that this association reflect the difference of longevity between reproductive and non-reproductive individuals, suggesting that reproductive status in eusocial or cooperative breeding is actually more correlated to longevity than sociality in itself (Lucas & Keller, 2020). Given the balance between the costs and benefits of social life, it might actually be hard to evidence association between sociality and patterns of senescence at the interspecific level.

³ Small species can also be social as it is the relative brain size which is the core idea in the social brain hypothesis. Small species with relatively big brain according to their body size should be social species according to this hypothesis. However, large species with larger brains should exhibit more complex social behaviours. Pérez-Barbería et al., (2007) insist however on the uncoupling of brain size and sociality that can be observed in ungulates or carnivores (Finarelli & Flynn, 2009) for example.

3.2. AT THE INTRASPECIFIC LEVEL

Social life affects the biological time of individuals. Indeed, social life may act on the rate of development and the age of first reproduction, or on the timing of reproduction in cooperative breeding species for example. In the meerkat, helped breeding females can reproduce more frequently and earlier in the season than less helped females (Russell et al., 2003) and helped meerkat pups had reduced age of first reproduction (Russell et al., 2007). In the dwarf mongooses (*Helogale parvula*) individuals attained reproductive status earlier when emigrating than in their natal groups (Rood, 1990). Living in groups can also reduce extrinsic mortality, as we discussed above, and lower rates of extrinsic mortality can be translated in longer developmental periods with more allocation of resources towards growth and somatic maintenance, consequently increasing longevity (Lucas & Keller, 2020). However, social behaviours vary in magnitude between individuals of a group and according to their recipient. The absence of evidence at the interspecific level is likely to be due to the fine intraspecific effects of the underlying costs of sociality on senescence.

Among groups, individuals may have to compete for breeding opportunities (Sharp & Clutton-Brock, 2011) and this intrasexual competition can result in lower reproductive success and/or survival and may alter rates of senescence in both males and females (Berger et al., 2018; Holekamp et al., 1996; Sharp & Clutton-Brock, 2011; Thorley et al., 2020). In cooperative breeding mammals, dominant breeding pairs monopolise the reproduction and subordinate individuals delay their own reproduction to provide alloparental care (i.e. care to offspring by non-parent individuals) to offspring (Koenig et al., 1992). This social organisation results in strong intrasexual competition to access dominant positions and thus reproductive status, and this has been shown to trigger lower reproductive success and higher rates of ageing in female meerkats undergoing strong competition (Sharp & Clutton-Brock, 2011). On the other hand, survival senescence of dominant meerkats of both sexes was absent (Thorley et al., 2020), and in adult dominant Alpine marmots (*Marmota marmota*) individuals helped during adulthood showed slower rates of ageing – although this pattern varied between sexes and life stages when helped (Berger et al., 2018) – evidencing also the benefits of social life on patterns of senescence.

Overall, patterns of senescence are modulated through social environment, especially during the early-life of individuals as the developmental period is a key period during which abiotic and biotic conditions can have long term consequences on fitness-related traits and senescence rates (Lemaître et al., 2015; Nussey et al., 2007). Indeed, environmental factors during early development play a primary role in explaining individual variations in lifespan or condition as

adult. For example, adverse conditions early in life in humans has been shown to result in impaired growth, reproductive success, survival, physiological functions or increased ageing (Lummaa & Clutton-Brock, 2002). These early effects can also be mediated through the social environment during development. In the domestic rabbit (*Oryctolagus cuniculus*), the position of pups within the litter huddle resulted in variations in body mass at weaning as more central individuals occupied the more energetically profitable position, and this difference in body mass could potentially have late-life fitness consequences (Zepeda et al., 2019).

In the following parts, we will develop the impact of social factors early in life on patterns of senescence using the predictions of two main hypotheses: the predictive adaptive response (PAR) hypothesis and the silver spoon effect. For this we will first place these hypotheses within the framework of the disposable soma theory. This theory will here offer an evolutionary context under which social factors early in life may affect patterns of senescence through the mechanisms described by the two former hypotheses.

3.2.1. Social development and the disposable soma theory

In brief, as described above, the disposable soma theory relies on the principle of the trade-offs in resource allocation between growth, reproduction and somatic maintenance as a consequence of extrinsic mortality and more generally environmental conditions (Kirkwood & Rose, 1991). The theory predicts that in adverse environmental condition, allocation towards reproduction should be prioritised, thus accelerating the rate of senescence (Kirkwood, 2005). For instance, young benefiting from alloparental care could save energy for growth and maintenance and show better survival and slower senescence.

In red wolves (*Canis rufus*), the presence of helpers providing care for pups affected mass and survival in contrasting ways according to the population density, with high densities resulting detrimentally for mass and with no effect on survival, while at low density both were positively affected by the presence of helpers. Female pups helped also presented longer reproductive lifetime and males showed reduced reproductive lifetime, whereas age at first reproduction was not impacted in both sexes (Sparkman et al., 2011). Prairie voles (*Microtus ochrogaster*) produced more frequent litters when reared with older siblings (Solomon, 1994). We can hypothesise that individuals reared with siblings could invest more heavily into reproduction at adult age because of decreased investment as pups in growth due to help from siblings. However, as specified earlier, costs of group-living include competition to access to food, reproduction or any other resources. In that regard, Nussey et al. (2007) have shown that females red deer born during year where population density was high, hence competition for resources was,

experienced higher rates of reproductive senescence as adults than when born during years of low population density. The two next sections will develop two hypotheses to explore mechanisms by which early-life conditions, and particularly social factors, may affect late-life fitness and patterns of senescence.

3.2.2. *The predictive adaptive response theory*

Phenotypic plasticity allows organisms to adapt to immediate changes in their environment thanks to the ability of one given genotype to produce several phenotypes (Via et al., 1995). Gluckman and Hanson (2004) exposed that this plasticity can occur during the early developmental period of an individual and can result in a permanent change during adulthood. However, they developed a second consequence of this plasticity, positing that early environmental conditions could induce changes in an individual that are expected to procure future advantages without immediate consequences. This is what they called predictive adaptive responses (PARs). In other words, environmental conditions during developmental periods may act as cues for future conditions in life, allowing individuals to adapt in advance to these future conditions, similar to the one experienced early in life (Figure 1a). For example, the meadow vole (*Microtus pennsylvanicus*) develops a thicker coat when born in autumn than in spring. This is induced at the pre-natal stage through mother cues. However this thicker coat does not provide any advantages to the young as intra-uterine and nest conditions should not differ between autumn and spring, however, in later conditions it will (Gluckman et al., 2005; Gluckman & Hanson, 2004)⁴. This leads to the consequence that an individual producing a predictive adaptive response to a poor-quality environment, should experience detrimental consequences (*i.e.* metabolic diseases) when placed under high quality, mismatching conditions (Figure 1a).

The PAR hypothesis has been documented in some cases. For example, cues of population density of North American red squirrels (*Tamiasciurus hudsonicus*), triggered hormonal responses by the mother inducing faster growth rate of offspring under high population densities, procuring advantages for the mother and the offspring by increasing the survival of the latter over its first winter (Dantzer et al., 2013). The author also found that faster growth in high population density conditions was advantageous for the offspring regarding its recruitment

⁴ Here lies the main difference between the PAR hypothesis developed in this review, and the thrifty phenotype hypothesis. Both are quite similar hypotheses but the former induces changes in an organism which advantages will not be perceptible until adulthood or more advanced age, while the latter induces permanent changes which advantages are directly observable, as adapted to current developmental conditions (Monaghan, 2008).

within the population. Sociality producing predictive adaptive responses has also been shown in Alpine marmots. Females senesced slower when social conditions (*i.e.* the presence or absence of helpers) were the same at birth and during adulthood, while when social conditions were not matching, females senesced faster (Berger et al., 2018).

However, still few studies assessed the contribution of sociality in the production of PARs. The PAR hypothesis is more widely used in medicine and evidence supporting it are more found in short-lived species (Dantzer et al., 2013; Gluckman & Hanson, 2004). Indeed, PARs are more likely to be produced in short-lived species as early- and late-life conditions are more likely to match than in long-lived species (Douhard et al., 2014).

3.2.3. *The silver spoon effects*

Several studies assessing the predictions of the PARs tended to find evidence against it, showing more support to the silver spoon effect (*e.g.* Douhard et al., 2014; Hayward & Lummaa, 2013). The silver spoon hypothesis posits that individuals developing in harsh conditions, would undergo fitness consequences in all kind of conditions later in life, good or poor, compared to individuals which developed in good early conditions. Developing in good conditions results in good quality individuals which fitness advantages are high in good adult conditions and decrease with deterioration of the environment. Poor-environment developing individuals will be of poor quality and should have higher fitness in good adult environments – decreasing with environmental deterioration – but will always perform less well than individuals born in good conditions, whatever the adult environment (Monaghan et al., 2008; Figure 1b). For example, banded mongooses (*Mungos mungos*) receiving care as pups, were heavier when reaching sexual maturity and experienced higher lifetime and reproductive success than unescorted pups (Vitikainen et al., 2019). Meerkats helped as pups had reduced age of first reproduction and were more likely to reach alpha rank (Russell et al., 2007).

4. STUDYING OF AGEING

4.1. MEASURING TRAITS AND MOSAIC AGEING

Social factors, notably during early developmental periods, may shape patterns of senescence through different mechanisms. However, we have seen in previous examples that senescence processes did not always act as a whole on an individual, but that sociality could impact differently survival or reproductive success. Instead, senescence is a general concept impacting individuals in asynchronous and heterogeneous ways. Senescence is a universal trait shared by

the (quasi-)entirety of the tree of life (Jones et al., 2014). However, patterns of senescence are highly diverse between species and between individuals, but also between traits within individuals, in part according to the shape and pace of senescence (Baudisch, 2011; Jones et al., 2014), and in other part according to its uniformity (*i.e.* synchronicity, homogeneity). This is observable through the large interindividual variation observed in ageing and senescing organisms, what may be referred as “differential ageing”. In their very poetic article, Walker and Herndon (2010) verbalised this phenomenon as “mosaic ageing”, defined by the variation in times and rates at which different changes occur in different systems of an organism. To be consistent with our definitions settled in preamble of this review, we will prefer to refer to “mosaic senescence” and “mosaic ageing” separately.

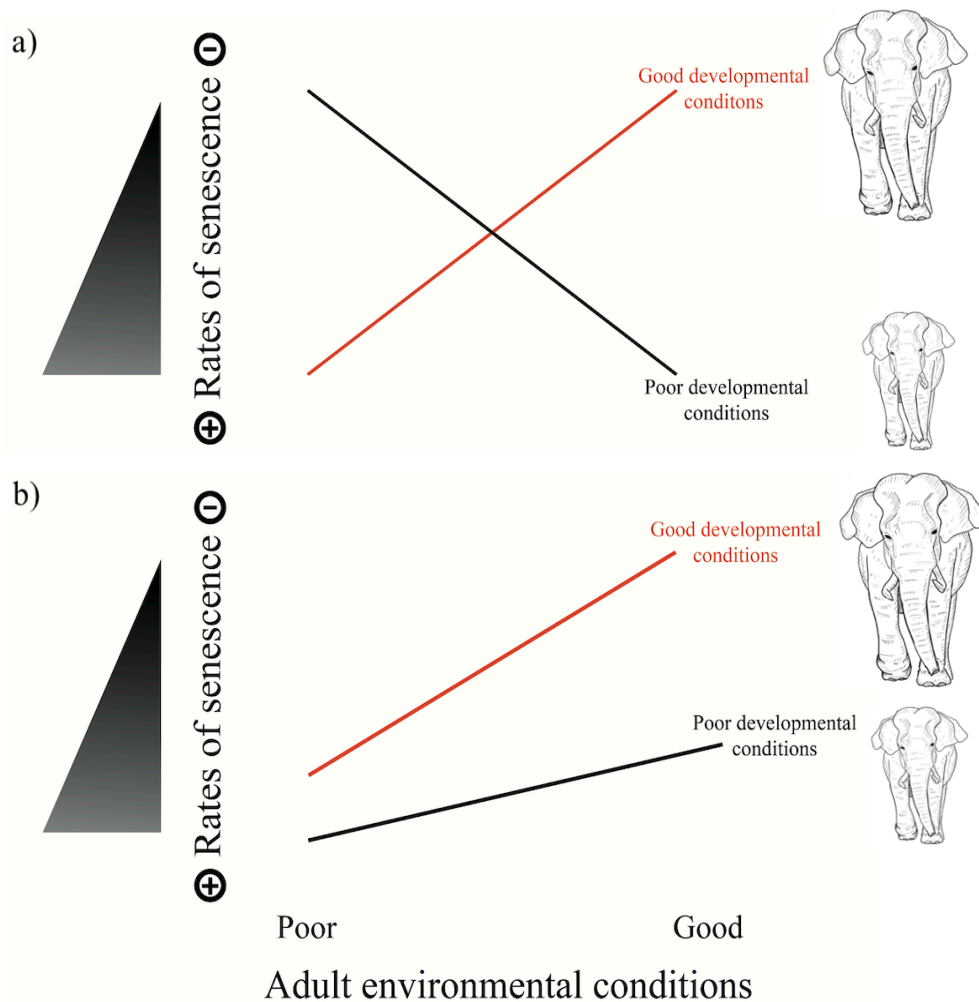


Figure 1. Predictions of the rates of senescence according to two hypotheses, depending on early- and late-life environmental conditions. The y axis grades from high rates of senescence to slow rates of senescence. a) Predictive adaptive response (PAR): individuals in adult conditions matching with their developmental conditions will perform better than individuals in mismatching environments, resulting in slower rates of senescence. b) Silver spoon effect: individuals developing in good conditions will always have slower rates of senescence than individuals developing in poor quality environments, but both individuals will perform better in good adult conditions (adapted from Monaghan, 2008).

To illustrate this, we can once again think about menopause. In addition to the fact that menopause occurs at different ages of individuals of a species experiencing it, it shows that reproductive senescence and senescence of other component of the organism affecting survival are uncoupled. Thus, it is common to talk about actuarial senescence when dealing specifically with an age-related decline in survival, reproductive senescence for an age-related decline in reproductive fitness or functional senescence to describe within-individual decline in physical and physiological functions with increasing age (Hämäläinen et al., 2014; Jones et al., 2008). This mosaic may explain some examples cited previously, like in dominant meerkats in which no survival senescence has been observed while reproductive senescence occurred (Thorley et al., 2020).

Due to this mosaic it is extremely difficult to study overall senescence in an organism. The study of senescence consists generally in the measure of a change in a trait value or decline of a function. This measure is expected to capture patterns of ageing of the trait under study. However, what is captured does not necessarily reflect a pattern of ageing as selective disappearance and lack of good quality data for advanced ages might be responsible for what is actually observed (Figure 2).

4.2. THE WITHIN- AND BETWEEN-INDIVIDUAL DIFFERENCES

One of the first issues arising when dealing with population of individuals varying in their traits is the one related to selective disappearance. Indeed, quantifying changes in a trait value within a population imposes to consider both within- and between-individual trait changes. When

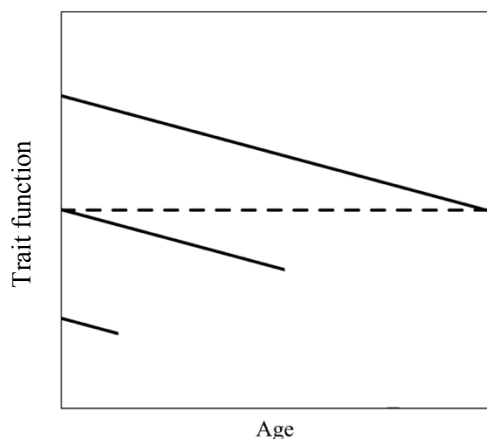


Figure 2. Senescence and selective disappearance of low-quality individuals. Age related decline in trait function for three individuals (solid lines) and observed pattern of ageing (dashed line) due to between- and within-individual variations (adapted from van de Pol and Verhulst, 2006).

observing a species at the group or population level, these existing between- and within-individual variation might actually mask patterns of ageing (van de Pol & Verhulst, 2006; Figure 2). As individuals with lower trait values should be counter-selected, *i.e.* reproduce less and/or survive less, it results that individuals of older age are mostly represented by individuals having a higher value or a better functioning of the trait under study. Moreover, those individuals have a trait value related to their age. Figure 2 describes the problem of selective disappearance when studying ageing. It represents three individuals

with different initial trait performance. Individuals with low trait function might be removed from the population with increasing age. The ageing direction of the trait (*i.e.* functioning decline, solid line) for the three individuals is masked by within- and between-individual variations in function decline (*i.e.* captured ageing direction, dashed-line).

Selective disappearance issues might also be revealed when not taking sex differences into account. Indeed, males and females usually experience different life histories, directly affecting patterns of senescence, and females, in mammals, tend to have longer lifespans than males (Bonduriansky et al., 2008; see also Box 1). In that regard, some studies explored patterns of senescence considering both interindividual variations and intersexual variations. For example, Hämäläinen et al. (2014) compared body mass ageing in a group of captive *versus* wild grey mouse lemurs (*Microcebus murinus*) considering both sex-effects and body quality-effects, allowing in one part to disentangle extrinsic and intrinsic mortalities, and in another part to assess the relative impact of both selective disappearance and functional ageing for the species studied. The authors found no evidence of functional ageing in the wild population – contrary to the predictions of the theories of ageing (*i.e.* higher rates of senescence associated with higher extrinsic mortality) – probably due to the strong removal of low-quality individuals through extrinsic mortality, evidencing selective disappearance effects. Another study showed a body mass-based selective disappearance, in females only, in three species of free-living ungulates (Nussey et al., 2011). Selective disappearance has been proposed as an explanation for studies showing no early-late life trade-offs and which did not take individual heterogeneity into considerations (Lemaître et al., 2015; Nussey et al., 2008b). Nussey et al. (2008b) advocate for the necessity of longitudinal studies in which individuals are recognizable and observed repeatedly to separate within- and between-individual variations and avoid selective disappearance issues or other issues such as the one arising when not considering the variations in resource acquisition and allocation between individuals and trade-offs between life history traits resulting in positive correlations (van Noordwijk & de Jong, 1986).

4.3. THE PROXIMATE ASPECTS OF SENESCENCE

In this section we will briefly give an example of a trait to measure in order to capture ageing and discuss about the two major proximate aspects explaining senescence, the oxidative damages and telomere attrition. Indeed, the regulation of the resource allocation towards this or that has to be explained physiologically by the regulation of the accumulation of damages within the organism. If oxidative damages and the telomere attrition processes are supposed mechanisms of senescence, the way these are regulated is still evasive.

4.3.1. *The example of the body condition*

Body condition in mammals is defined by the energetic state of an individual, *i.e.* the amount of energy reserves, usually fat (Schulte-Hostedde et al., 2001). A good body condition is associated with higher reproductive success or longer reproductive lifespan in females or competition success between males (Schulte-Hostedde et al., 2001). Body condition can be approximated by morphological measurement of body mass linked with morphological measurements such as body size or chest girth. Body mass can be interpreted as a measure of the resources that can be allocated to different systems of an organism (Hämäläinen et al., 2014). In that respect, body mass is a good way to study functional senescence of an individual. Moreover, body mass is a highly repeatable trait over one's own lifetime (Briga et al., 2019), and predicts survival and/or reproductive success in many species (Briga et al., 2019; Festa-Bianchet et al., 1998; Hämäläinen et al., 2014; Monclús et al., 2014; Nussey et al., 2011; Zepeda et al., 2019). Moreover, we have seen in previous examples that body mass may also be affected by social environment, especially during early-life, as with domestic rabbit pups in a central position within the huddle were heavier for example (Zepeda et al., 2019) or with the presence of older siblings during the rearing period of prairie voles (Solomon, 1994). All of these criteria give body mass good credits to focus on in order to study patterns of senescence in wild populations.

4.3.2. *Oxidative stress*

The free radical theory relies on the production of reactive oxygen species as a by-product of the production of ATP by the mitochondria. Those highly reactive molecules can then accumulate and result in oxidative damage to proteins, DNA and other components of the organism (Monaghan et al., 2008). Oxidative damages, as a measure of oxidative stress, are expected to be the result of the balance between the production of free radicals (oxidants) and of antioxidative molecules (antioxidants) (Costantini & Verhulst, 2009). As the oxidative damages are a good candidate to explain age-related deterioration of an organism, it seems relevant to suppose that mechanisms exist to regulate this accumulation of reactive molecules in order to slow the rate of senescence when extrinsic mortality is low for example. Lucas and Keller (2020) reported studies showing higher levels of oxidative stress in naked mole rats than mice while the former are considerably much longer-lived. However, naked mole rats experienced slower increase of these molecules and higher protein translation fidelity.

In Alpine marmots, individuals from larger groups experienced lower oxidative damages, potentially resulting from benefits of group-size for social thermoregulation during hibernation

(Lardy et al., 2016). Nevertheless, Lucas and Keller (2020) reported in their review that evidence for such a mechanism are still scarce⁵.

4.3.3. *Telomere attrition*

Telomeres are protective non-coding sequences of DNA ending chromosomes. As replications repeat, these telomeres are shortened down to a point where replication ceases and the cell is destroyed or remains within tissues. In that last case, the cell produces damaging substances that accumulate over lifetime (Monaghan et al., 2008). It had been found that telomeres attrition was quicker early in life rather than at adulthood (Brümmendorf et al., 2002) and tended to be associated with growth rates, highlighting potential influences of telomere attrition in the trade-offs between growth, reproduction, survival and maintenance and thus patterns of senescence (Monaghan et al., 2008). In spotted hyenas (*Crocuta crocuta*), higher-ranked individuals exhibited longer mean telomeres length than subordinates, suggesting that social ranking might contributes to senescence (Lewin et al., 2015). However, once again, the study of telomere attrition to quantify senescence might be misleading. Indeed, Cram et al. (2018) have shown that dominant meerkats, which outlive subordinates, had actually higher rates of telomere attrition than the latter ones. As mentioned above, the observed pattern in which dominant live longer than subordinates seemed to be explained by the higher extrinsic mortality of subordinates with age – due to increased dispersal with age – misinterpreted as senescence. More research is needed concerning the way the proximate aspects of senescence and ageing are regulated according to environmental and social conditions.

5. DISCUSSION AND PERSPECTIVES

Throughout this review we have explored how a paradigm shift had occurred between the 1930's and 1950's through the insight that the strength of the natural selection was supposed to decrease with advanced age, and, as a consequence, that early-life advantages should undergo a selective premium, compared to the sorting out of late-life disadvantages. This has led to different hypotheses, constituting the classical evolutionary theory of ageing. We then explained the need of extending this theory by including social components into it, affecting patterns of senescence through kin-selection theory and group-living costs and benefits. We

⁵ Recent studies have found that oxidative damages were actually not associated with age but with 'frailty' (*i.e.* a greater vulnerability to minor stresses). In this theory, senescing in itself is not associated with oxidative damages, but unsuccessful senescence is (Viña, 2019; Viña et al., 2018). We might expect no association between oxidative damages and unsuccessful senescence in wild animals, as these frail individuals should be intrinsically more susceptible to extrinsic mortality.

emphasised the importance of early-life developmental conditions on senescence within the framework of the disposable soma theory and highlighted that the silver spoon effect and the predictive adaptive responses could help predict rates of senescence, although these effects are debated. We highlighted that senescence was a mosaic process resulting in within- and between-individual variations and that only longitudinal studies were able to deal with these variations. We also underlined the need for a better understanding of the mechanisms regulating the physiological processes explaining ageing and senescence. This review allowed us to apprehend the already existing knowledge concerning the relation between social factors and patterns of senescence, and the gaps existing in our comprehension of this relation.

In that regard, we aim now at introducing a future research project contributing in filling these gaps. This research project will investigate the link between early-life social environment and body mass ageing in semi-captive population of Asian elephants (*Elephas maximus*). We will use longitudinal data of a semi captive population in Myanmar, used as workers for wood logging. Through repeated measurements of the same individuals – to avoid selective disappearance issues – and based on the above theories we will here test three predictions:

(i) the presence of elder siblings during the developmental period of a focal individual is expected to result in slower rates of body condition deterioration because younger siblings should increase sibling competition for resources, (ii) individuals which mother was absent during development should suffer earlier and higher rates of body condition deterioration, (iii) young individuals able to interact with aunts are expected to deteriorate later and/or slower due to potential allomaternal care by aunts. Body condition ageing trajectory will be estimated through morphological measurements (body mass, chest girth or body length). We might also expect sex-differences of body condition deterioration depending on social factors. Indeed, due to the existence of grandmother effects in Asian elephants, we might expect females to undergo delayed and/or slower rates of ageing in order to provide longer alloparental care (Lahdenperä et al., 2016; Natrass et al., 2019). On the other hand, Asian elephants living in matriarchal social structure may be more subject to competition for higher hierarchical position or food resources, while males disperse and are mainly solitary (Fernando & Lande, 2000).

This research project aims at filling the gap surrounding the relation between social factors and patterns of senescence and more precisely, about the early-late life trade-off linking sociality and senescence in long-lived highly social species. Furthermore, this research might provide insight in the management of the semi-captive population of Myanmar and could thus be of conservation interest, by preventing the need for depleting natural wild populations of Asian elephants to supplement the semi-captive one.

6. ACKNOWLEDGEMENTS

I would like to thank my two supervisors, Dr. V  rane Berger and Dr. Michael Briga, for their many helps and advises, both for me to complete this review and for my so far successful settlement in Finland. I am also thankful to my two brothers, Valentin Lalande for putting the first figure in this review in a nice, readable way, and Daniel Lalande for the beautiful Asian elephant drawing. If this review is given a bad mark, however, know that I will come back to haunt each one of you and will put tiny rocks in your shoes every morning.

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8. SUMMARY

The first evolutionary theories of ageing emerged in the 1950's thanks to the insight that the strength of selection should weaken with advanced ages because of extrinsic mortality while senescence was at first restricted to humans, domestic and captive animals. However, these classical theories of ageing are currently unable to provide a satisfactory explanation of the huge diversity of patterns of senescence that we observe in the wild such as long postreproductive lifespans or the existence of non-reproductive individuals in cooperative breeding species for example. In the recent years, attempts to include sociality in the theories of ageing seemed able to provide a more comprehensive view of this diversity through kin selection theory or the fact that social life can result in a decreased extrinsic mortality affecting patterns of senescence by delaying or slowing rates of senescence. Evidence of an association between longevity and sociality are scarce at the interspecific level probably due to the fine effects of sociality at the intraspecific level and the balance between costs and benefits of life in society. At the intraspecific level a more visible association between sociality and patterns of senescence emerges through the link between sociality and pace of life, cooperation and competition between kin, decreased extrinsic mortality or effects of social environment during early-life. However, a comprehensive view of senescence needs to incorporate the mosaic nature of senescence. The study of senescence implies the need for longitudinal studies with repeated measures of the same individuals to avoid methodological issues such as selective disappearance. It is therefore important to focus also on the proximate causes of senescence and their regulations at the physiological level, shown to be associated with social factors in some cases, but few studies attempted to do so yet. In that regard our research project aims at filling the gap about the early- and late-life trade-off linking social environment and ageing in long-lived species. We will use longitudinal data to study the effects of different components of the early social life on the trajectory of the body condition ageing in a semi-captive Asian elephant (*Elephas maximus*) population in Myanmar.